

Emanuele Bugliarello

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Research Interests

Multimodal and multilingual natural language understanding, representation learning, transfer learning

Education

PhD in Computer Science
University of Copenhagen
Advisors: Desmond Elliott, Anders Søgaard
2019–present
Copenhagen, Denmark

MSc in Communication Systems, Data Analytics specialisation
École Polytechnique Fédérale de Lausanne (EPFL)
Thesis: *Matrix Completion in the Unit Hypercube via Structured Matrix Factorization*
GPA: 5.53/6 (3/44)
2015–2018
Lausanne, Switzerland

BEng in Electronics and Information Engineering
Tongji University (同济大学)
Thesis: *General large-batch methods for scalable accurate training of neural networks*
GPA: 29.34/30
2013–2018
Shanghai, China

BSc in Telecommunications Engineering
Politecnico di Torino
GPA: 28.86/30 (3/120) – Graduation Grade: 110/110 *Cum Laude*
2012–2015
Turin, Italy

Machine Learning Schools: LxMLS 2020, Oxml 2020, ALPS 2021

Experience

Visiting Scholar with Prof. Siva Reddy
Mila
01/2022–present
Montreal, Canada

Research Scientist Intern with Rishabh Mehrotra and Mounia Lalmas
Spotify Research
On *Multi-Objective Music Sequencing via Counterfactual Beam Search*
08/2020–11/2020
London, UK

Research Associate with Prof. Naoaki Okazaki
Okazaki Lab @ Tokyo Institute of Technology (東京工業大学)
On *Neural Machine Translation: Dependency-aware Self-Attention and Cross-Mutual Information*
10/2018–09/2019
Tokyo, Japan

Visiting Scholar with Prof. Rui Fan and Prof. Dan Alistarh
ShanghaiTech University (上海科技大学)
On *Large-Batch Algorithms for Deep Learning at Scale*
04/2018–07/2018
Shanghai, China

Research Intern with Swayambhoo Jain, Brian Eriksson and Jean Bolot
Technicolor AI Lab
On *Intelligent Visual Effects Scheduling via Constrained Matrix Factorization*
09/2017–03/2018
Palo Alto, CA, USA

Research Assistant with Marcel Schöngens, Maxime Martinasso and Claudio Gheller
Swiss National Supercomputing Centre (CSCS)
On *Distributed TensorFlow: A performance evaluation*
06/2017–09/2017
Lugano, Switzerland

Teaching

Teaching Assistant, EPFL: Internet Analytics (COM-308)
Teaching Assistant, UCPH: Advanced Topics in Natural Language Processing (NDAK19001U)
Teaching Assistant, UCPH: Neural Information Retrieval (NDAK20002U)
Spring 2017
Winter 2020
Spring 2021

Publications

8. **On Language Models for Creoles**
H. Lent, E. Bugliarello, M. de Lhoneux, C. Qiu, A. Søgaard
In *CoNLL 2021*
7. **Visually Grounded Reasoning across Languages and Cultures**
F. Liu*, E. Bugliarello*, EM. Ponti, S. Reddy, N. Collier, D. Elliott
In *EMNLP 2021* **Best Long Paper Award**
6. **Vision-and-Language or Vision-for-Language? On Cross-Modal Influence in Multimodal Transformers**
S. Frank*, E. Bugliarello*, D. Elliott
In *EMNLP 2021*
5. **Multimodal Pretraining Unmasked: A Meta-Analysis and a Unified Framework of Vision-and-Language BERTs**
E. Bugliarello, R. Cotterell, N. Okazaki, D. Elliott
TACL 2021
4. **The Role of Syntactic Planning in Compositional Image Captioning**
E. Bugliarello, D. Elliott
In *EACL 2021*
3. **It's Easier to Translate *out of* English than *into* it: Measuring Neural Translation Difficulty by Cross-Mutual Information**
E. Bugliarello, SJ. Mielke, A. Anastasopoulos, R. Cotterell, N. Okazaki
In *ACL 2020*
2. **Enhancing Machine Translation with Dependency-Aware Self-Attention**
E. Bugliarello, N. Okazaki
In *ACL 2020*
1. **Matrix Completion in the Unit Hypercube via Structured Matrix Factorization**
E. Bugliarello, S. Jain, V. Rakesh
In *IJCAI 2019*

Invited Talks

- **Visually Grounded Reasoning across Languages and Cultures** CIMEC, University of Trento 06/2021
Samsung AI Center - Toronto 11/2021

Skills

Programming

- Python, Matlab
- NumPy, Pandas, scikit-learn
- PyTorch
- CUDA, SQL

Languages

- native: Italian
- fluent: English (TOEFL iBT 109)
- intermediate: French
- basic: Chinese (HSK III), Japanese, Danish

Fellowships, Honors, and Awards

EuroHPC Grant with Desmond Elliott on "Vision and Language on a Shoestring without Tear"	06/2021
University of Copenhagen Marie Skłodowska-Curie TALENT Doctoral Fellowship	06/2019
EPFL PhD Program in Computer Science Fellowship [offer declined]	05/2019
The Japanese Government (MEXT) Scholarship [offer declined]	01/2019
PoliTong Merit Scholarship	07/2013
MIUR National Register of Excellences	06/2012

Service

Reviewer: ARR, EMNLP, ICLR, IJCAI, NeurIPS

Organiser: Multilingual Multimodal Learning Workshop (ACL 2022)